Imagining Risk Taking: The Valence of Mental Imagery Is Related to the Declared Willingness to Take Risky Actions

Running title: Mental Imagery and Risk Propensity

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Abstract

The aim of the present research was to investigate the involvement of mental imagery in people's choices under risk. We tested the general idea that decision makers can use visual mental images (visual mental simulations) to pre-experience how rewarding or threatening future outcomes of risky behavior will be and try out the potential consequences of their risky activities. The paper reports the results of three preregistered studies (including one experiment) showing that the valence of mental imagery is related to the willingness to take risky actions and that people spontaneously use mental imagery as an informative decision input. In Study 1, we found that the more positive mental images people produced when faced with risk, the more willing they were to take risky actions representing different risk domains. Study 2 extended the results of Study 1, indicating that the valence of mental imagery has a causal effect on participants' risk-taking willingness. Qualitative analysis based on independent judges' evaluations conducted in Studies 1 and 2 documented that, when requested, participants could easily generate visual mental images illustrating the consequences of their risky choices. Finally, with Study 3, we found that participants declared using mental imagery as a decision input (i.e., a source of information that helps them make choices) even when they were not instructed to do so. However, the frequency of reporting images as decision inputs differed across risky activities.

Keywords: mental imagery, risk taking, willingness to take risks, risky actions

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Introduction

Decision making is inextricably associated with thinking about the future as part of a broader process of mental time travel (Boyer, 2008; Suddendorf & Corballis, 2007; Szpunar, 2010; Zaleskiewicz, Traczyk et al., 2023). An essential aspect of decision-related information processing directly linked to episodic foresight (Suddendorf, 2017) is the consideration of potential future outcomes of current choices - their type, magnitude, and valence (Beach, 2009, 2016; Tuckett & Nikolic, 2017). For example, a decision maker may think about the possibility of a future loss but underestimates its extent. Such a misevaluation can lead to choosing an option that is inconsistent with the decision maker's preferences and exceeds the limit of personally acceptable risk. The present research shows and discusses the findings from three preregistered studies (including one experiment) testing the general prediction that mental imagery may be a critical psychological mechanism used to simulate potential future outcomes of risky activities. In particular, we show that generating and processing more positive mental images of risky actions is associated with higher risk acceptance while creating negative mental images makes a decision maker more risk averse. We also provide preliminary evidence that decision makers use mental imagery as informational input spontaneously, even if they are not encouraged to do so.

Mental Imagery and Decision Making

Before investigating how mental imagery might inform decision making, the concept itself should be explained. Cognitive psychology offers a broad spectrum of definitions. The most widely cited ones were introduced by Kosslyn et al. (2001) and Pearson et al. (2015). The former authors proposed that "mental imagery occurs when perceptual information is

accessed from memory, giving rise to the experience of 'seeing with the mind's eye,' 'hearing with the mind's ear' and so on…" (p. 635). For the latter authors, mental imagery comprises "representations and the accompanying experience of sensory information without a direct external stimulus" (p. 590). Both definitions are useful when studying the association between mental imagery and risk taking. When making current choices, people do not experience their outcomes immediately, but they can simulate them in their imagination. In the present instance, mental imagery is interpreted as a cognitive tool that allows an individual to simulate the future (Beach, 2009; Nanay, 2016, 2023). We also refer to the concept of mental emulation as propounded by Moulton and Kosslyn (2009). They argued that the primary function of mental imagery is to generate predictions; in particular, it allows us "to predict what we would experience in a specific situation or after we perform a specific action" and "to anticipate what may occur in the near and distant future" (p. 1273). They gave an example to illustrate how mental images of a particular behavior can produce simulations of an unknown (future) experience: "To simulate a conversation via emulation, for example, you could place yourself in the 'mental shoes' of those conversing, predicting their dialogue based upon how you would respond (based on your emotions and the associations that are triggered) in their respective situations" (p. 1276). According to such an interpretation, mental imagery can be used to *pre-experience* how rewarding one's engagement in a given risky activity would be (Blackwell, 2020) and *try out* (both cognitively and emotionally) the potential consequences (Atance & O'Neill, 2001; Schacter et al., 2017). It should be noted that mental imagery can occur in different sensory modes (e.g., the auditory, haptic, gustatory, and olfactory modes), but in our present research, we focused on the specific role played by the visual mode (single images or series of images in the form of 'mental films'). Previous research has shown that people report greater vividness for vision than for other modalities, which suggests that the visual modality plays a dominant role in mental imagery (Belardinelli

et al., 2004; Schifferstein, 2009; Switras, 1978). Thus, we propose that when people use mental images in their decision making, they are more likely to generate visualizations rather than images based on other modalities.

Basing one's decisions of whether to engage in risky actions on mental imagery seems highly adaptive (Zaleskiewicz, Traczyk et al., 2023). If an individual can relatively quickly generate mental images that portray the negative consequences of risk taking, their willingness to accept risky options should diminish. This, in turn, might protect them against exceeding the limits of acceptable uncertainty. Such reasoning is supported by the fact that mental imagery is closely related to emotions (Blackwell, 2020; Holmes & Mathews, 2010; Ji et al., 2016; Nanay, 2023). For example, generating powerful negative images of a threat would evoke a strong negative affect, which will discourage the decision maker from engaging in excessive risk taking.

The idea that people can use mental simulations to recognize the shape of future actions was earlier developed in the form of a simulation heuristic (Kahneman & Tversky, 1982). MacLeod (2017) proposed that the simulation heuristic can be seen as "the process of constructing a mental model of reality in which the hypothetical model takes place" (p. 273). He also postulated that when people engage in mental simulations, they actively construe causal explanations of events rather than just retrieving past experiences. Kappes and Morewedge (2016) suggested that when people mentally simulate evidence, they tend to use it as a substitute for physical evidence, with the former having a disproportionate effect on estimations of the probability of future events. In their view, "[s]imulations affect expectations about the future because, like actual experience, they are interpreted as providing evidence about why and how events will actually occur" (p. 409). Evidence supporting such reasoning indicated that people rated the likelihood of future adverse events (e.g., developing a physical illness) as higher when they engaged more intensively in mental simulations and

when they could more easily create visual images of those events (Raune et al., 2005). We would argue that mental imagery can be used as a form of mental simulation, allowing the decision maker to visualize future outcomes of their actions that (depending on their valence) can affect one's willingness to accept risk.

Previous Research Linking Decision Making With Mental Imagery

Evidence pointing to the role of mental imagery in decision making emerged in areas such as intertemporal and moral decision making and risk perception. When people are instructed to imagine future (delayed) outcomes, they become more prone to hold back from impulsively choosing immediate rewards. In Peters' and Büchel's (2010) study, participants made choices between monetary rewards that differed in both amount and time. Those who were cued to engage in episodic foresight shifted their preferences towards a delayed outcome. Meanwhile, Bulley et al. (2019) instructed participants to mentally simulate either negative or positive future scenarios. They found that people in both experimental conditions more often chose larger and delayed rewards than those in the control condition. In sum, it seems that using mental imagery to simulate delayed outcomes supports self-control in intertemporal decision making.

Research on moral judgment and choice has shown that visualizing immoral behaviors with closed eyes (which provokes engagement in mental simulations and evokes more intense emotions) led to a greater condemnation of such behaviors (Caruso & Gino, 2011). Amit and Greene (2012) found that visual mental imagery supports deontological moral judgments (i.e., those favoring the rights of the individual) over utilitarian moral judgments (i.e., those favoring the greater good). In particular, the authors observed that people with a more visual cognitive style tended to make deontological rather than utilitarian moral judgments and that introducing visual interference made moral judgments less deontological and more utilitarian.

The implication here is that mental imagery can regulate moral decision making and make people more likely to follow deontological ethics.

There is some evidence of a connection between mental imagery and risk perception. For example, one of the reasons behind underestimating the risk of climate change is that people generate amorphous and unvivid mental images of threats that climate change might cause. Leiserowitz (2006) asked participants to report mental images that appeared in their minds when thinking about global warming and to rate these images on a scale from *very negative* to *very positive*. The participants generated mental images that were not personally relevant (e.g., they included distant locations or affected nonhuman species). Those images evoked moderate or weak affective reactions (see also Leiserowitz, 2006; Smith & Leiserowitz, 2012). Recent studies on people's perception of risks related to COVID-19 have shown that instructing people to generate mental images of the negative consequences of infection made subjective risk assessments more accurate (Sinclair et al., 2021).

Finally, studies that used measures based on eliciting verbal imagery (Peters & Slovic, 1996) also documented that the perception of threats can be based on processing images. It has been found, for example, that people's support for nuclear power was related to the valence of images (defined as associations or thoughts) they generated (Peters & Slovic, 1996; Slovic, Layman, et al., 1991; Slovic, Flynn, et al., 1991). Benthin et al. (1995) used the word association technique to investigate the role of imagery in adolescent risk-taking behaviors. These authors found that health-threatening behaviors had many positive associations/images (e.g., having fun), which might explain why young people are exceptionally ready to engage in high-risk activities.

In sum, the results of previous studies suggest that mental imagery is associated with different aspects of decision making (see also Zaleskiewicz, Traczyk et al., 2023). The main purpose of the present project was to investigate the relationship between the valence of

visual mental images and people's willingness to accept risky options. The second was to examine whether people declare a spontaneous use of mental imagery in their decision making.

Overview and Contribution of the Present Studies

We report the results of three preregistered studies in which we focused on investigating the issue of whether producing mental images of risky activities is related to the willingness to take them. In Study 1, people were presented with risky activities and instructed to generate up to three mental images that crossed their minds when visualizing their involvement. They later rated the valence of their images and declared their willingness to engage in a given risky action. We found that the valence of mental imagery positively correlated with the readiness to take risky actions. In Study 2 (an experiment), we documented a causal link between the positivity/negativity of visual mental images and declared willingness to take risks. Participants who were instructed to generate positive mental images declared a greater readiness than the control group to engage in risky behavior. By contrast, those who generated negative images were less willing to accept risk than the control group. Finally, Study 3 showed that envisioning the future through imagination served to assist decision making. Participants stated that they used spontaneous mental images as decision inputs.

We believe that our present research contributes to the literature on how mental imagery is associated with risk taking in several ways. First, we specifically focused on the role of visual mental imagery related to considering risky actions. Instructions that were given to participants requested them to visually imagine risky actions as if they would see them via their 'mind's eye' (i.e., in the form of single images, series of 'mental pictures', or 'mental films'). Previous research investigating risk-related imagery (Benthin et al., 1995; Peters & Slovic, 1996; Slovic, Layman, et al., 1991; MacGregor et al., 2000; Slovic, Flynn, et al., 1991) mainly used the methods of 'continuous associations' or 'thought-listing' procedures

and, in some cases, instructed participants to think about risk instead of imagine or visualize it. In such methods, the participant's task is to provide associations that come to their minds in relation to a stimulus. The associations typically take the form of thoughts (words or phrases), meaning participants are not requested to generate visual mental images. It is possible that even though these studies employed 'thought-listing' procedures, participants might have also generated visual images. Amit et al. (2017) demonstrated that when people are prompted to think verbally and use inner speech, visual imagery is also involved to some extent. However, the authors of previous studies examining risk-related imagery did not report such data. In summary, our procedure aimed to activate and engage visual processing, with a primary focus on investigating visual mental imagery (Wyer et al., 2008).

Second, in two out of three studies we report in this paper, we run qualitative analyses in which independent judges evaluated whether participants indeed generated visual images. The aim of this procedure was to make sure that the effects we report concern visual mental imagery and not verbal associations people may also produce when faced with risk.

Third, in Study 3, we investigated the degree to which people can spontaneously generate mental imagery when considering their engagement in risky actions. To the best of our knowledge, in the vast majority of studies examining the role of mental imagery in decision making, participants were explicitly instructed to generate images. This suggests that prior research, unlike our Study 3, tested enforced rather than spontaneous use of mental imagery in decision-related contexts.

Study 1

Study 1 investigated whether the valence of mental images was significantly related to participants' declared willingness to take risky actions. We expected that there would be a positive correlation between the valence of mental images and the willingness to take action:

the more positive the ratings of mental images generated by participants, the higher their declared willingness to engage in risky actions would be.

Method

Participants

The sample size for this study was computed with a priori analysis using G*Power (Faul et al., 2009), which revealed that given an alpha of .05 and power of .95, a sample size of 115 participants would be sufficient to observe a moderate effect size (r = .30). Having in mind that some participants in online studies withdraw or do not pass attention checks, we decided to increase this number and recruit 230 participants from the United States via the Internet platform Prolific (122 females, 108 males, M_{age} = 38.93, SD = 13.09). All of them received remuneration ($1.64). In line with preregistration, responses provided by nine participants who completed the study using mobile phones were excluded from the analysis. Participants provided informed consent after they had been briefly informed about the general purpose of the study and the voluntary nature of their involvement. They were also told that they could withdraw at any time. The study was designed and conducted in accordance with guidelines enforced by University Ethical Committee and preregistered at the Open Science Framework (https://doi.org/10.17605/OSF.IO/JDKZX).

Procedure

The study was conducted online. In its respective phases, participants read brief descriptions of different real-life risky activities (e.g., “Taking strong medications without consulting your doctor” for the health domain; see Table 1 for a full list). These were displayed in random order. The participants’ task was to vividly imagine themselves being actively involved, that is, seeing the scene in their mind’s eye and describing up to three of the first mental images. They then rated the valence of each image (“Was the image positive, negative, or neutral?”) using a 100-point slider ranging from 0 - *very negative* to 100 - *very*

positive. Finally, for each risky activity, they rated their willingness to take a risky action ("Would you be willing to take this action?") on a scale ranging from 1- *definitely no* to 7 - *definitely yes*.

Results

Qualitative Analysis of the Content of Entries Presented by Participants

Before testing our main hypothesis, we conducted a non-preregistered, qualitative content analysis of the descriptions (entries) provided by participants. The purpose was to examine whether participants followed the instruction and created and described visual images related to the risky activities they were presented with. First, two independent judges fluent in English read the content of the entries and assigned each of them separately to one of two deductive codes - "visual imagery" and "non-imagery." (See Supplementary Materials for the exact wording of the codes' definitions.) Second, the third judge resolved any disagreements that arose in assigning each entry to one of the two codes by making a final, independent decision. To ensure high fidelity of analysis, each judge completed theoretical and practical training (1.5 h in total) before starting the task. The analysis was conducted using the MAXQDA software (Kuckartz & Rädiker, 2019).

First, we checked how often participants provided all three entries when reporting their mental images (96%). We collected 3281 entries in total. Independent judges agreed in their content analysis that 2461 (75%) of participants' entries described mental images. Among the entries assigned to the "visual imagery" code, participants provided descriptions such as: "I am picturing my husband being at work and needing to sign something for him," "I imagined being evicted from my house," or "I visualized the mountains of Arizona as I sped through them recklessly. Beautiful and tall with a sunrise." (See Supplementary Materials for additional examples of entries provided by participants in Study 1).

Hypothesis Testing

For the main preregistered analysis, responses were averaged across all risky activities for each participant separately. Next, we formed the general indices of the mean valence of reported mental images (M = 32.148, SD = 15.029, Cronbach's α = .756) and the mean willingness to take a risky action (M = 2.645, SD = 1.255, Cronbach's α = .646). Descriptive statistics for each specific activity are shown in Table 1. In the first step of the analysis (in line with our preregistration), we used responses provided by all participants without excluding entries evaluated by independent judges as belonging to the "non-imagery" category.

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Table 1

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First, we investigated the mean valence of mental images generated by participants and found that it was significantly lower than 50: $t(220) = -17.659$, $p < .001$, Cohen's $d = -1.188$, 95% CI [-1.359, -1.015]. A parallel analysis was conducted for the mean willingness to take risk (the dependent variable): $t(220) = -10.123$, $p < .001$, Cohen's $d = -0.681$, 95% CI [-0.822, -0.534]. This suggested that participants, when confronted with risky situations, tended to produce negative rather than positive mental images and preferred to refrain from taking risky actions.

Second, we examined the relationship between the mean valence of the mental images and the mean willingness to take risky actions. In line with our preregistered hypothesis, we observed a moderate and positive correlation between the two variables: $r(219) = .543$, $p < .001$, 95% CI [.459, 1.000]. This suggested that the more participants tended to evaluate their mental images as positive, the more willing they were to take risks.

To strengthen the support for our hypothesis, we conducted an additional, non-preregistered analysis of the valence of mental images and the willingness to take risky action using only those entries that were coded as "visual imagery" in the qualitative analysis. Before conducting the analyses, we excluded 820 participants' entries and data units related to them (i.e., ratings of the valence of mental imagery and ratings of willingness to take risks), which were rated by the independent judges as non-images. Five participants did not provide any instance of visual mental imagery, and their responses were entirely excluded from this analysis. Next, we averaged the remaining responses for each participant separately to create the general indices of the mean valence of mental images and the mean willingness to take risky actions. The quantitative analyses were conducted using the general indices. We examined the mean valence of the mental images and again found that it was lower than 50: $t(215) = -15.709$, $p < .001$, Cohen's $d = -1.069$, 95% CI [-1.235, -0.901]. The mean willingness to take risky action also fell below the middle of the scale: $t(215) = -10.308$, $p < .001$, Cohen's $d = -0.701$, 95% CI [-0.850, -0.552]. Finally, the correlation between the mean valence of the mental images and the mean declared willingness to take risks was again moderate and positive: $r(214) = .506$, $p < .001$, 95% CI [.418, 1.000].

To address potential variability in ratings of both valence and willingness to take risks across the five risky activities, we fitted a linear multilevel regression model with participants and activities as random intercept effects. Additionally, we allowed the effect (slope) of valence on willingness to take risk to vary depending on risky activity. Models were fitted using the lme4 package (Bates et al., 2014, p. 4) run in R statistical environment (R Core Team, 2022); p-values were estimated using the lmerTest package (Kuznetsova et al., 2014), while all pairwise comparisons were performed with the emmeans (Lenth, 2023) and ggeffects (Lüdecke, 2018) packages.

We corroborated the results from preregistered analyses reported earlier in this section. In particular, the mean valence of mental images generated by participants was significantly lower than the middle of the scale (i.e., 50): b = -17.853, t(5.515) = -4.955, p = .003, δt[1] = -0.707 (Fig. 1A). The similar effect was observed for the mean willingness to take risks: b = -1.355, t(5.490) = -4.281, p = .006, δt = -0.655 (Fig. 1B). Results did not change significantly when only records categorized as visual mental images were taken into account. Additionally, we found nonnegligible variability across risky activities employed in the study (for descriptive statistics aggregated on the participants level, see Table 1), with risky activity in the financial domain demonstrating the lowest mean willingness to engage (Fig. 1B) and the weakest relationship between the valence of mental images and the willingness to take risk (Fig. 1C). Despite of this, the fixed effect of the valence of mental images on risk taking was still positive and significant, b = 0.044, p < .001, 95% CI [0.030, 0.058], δt = 0.028, indicating that the type of a risky activity did not impact the overall pattern of results.

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Figure 1

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Summary

When instructed to do so, participants could easily produce mental images related to their involvement in risky actions, but these images tended to be negative rather than positive. From the perspective of our preregistered expectations and the general assumptions underpinning our theoretical model, the mean valence of mental images was positively correlated with the mean declared willingness to take risks. These results held after excluding data units (i.e., ratings of the valence of mental images and the willingness to take risky

[1] δt is a measure of effect size calculated by dividing the beta regression parameter by the square root of the sum of all residual variances.

action) related to those participants' entries that were classified as non-images by three independent judges and after accounting for variability across risky activities. In other words, the more positively participants could see their involvement in a risky activity through their mind's eyes, the more ready they were to take action. The results of Study 1 supported our hypothesis that mental images would increase the willingness to take risks when they were positive and vice versa.

Study 2

In Study 1, we demonstrated that the valence of mental imagery is correlated with the readiness to take risks. The purpose of Study 2 was to show that both variables were causally related; that is, the valence of mental imagery would influence risk proneness. Based on the correlations that were found in Study 1, we assumed that imagining positive circumstances around risky actions would lead to a higher propensity to take risks, whereas generating negative mental images relating to a risky action would have the opposite effect. Additionally, we attempted to replicate the correlational findings observed in the first study, again hypothesizing that the valence of mental images rated by participants would be positively correlated with the willingness to take risky actions.

Method

Participants

While computing the sample size, we used a commonly assumed medium effect size (Cohen's $f = .25$) for the expected difference between the three experimental groups. A priori analysis using G*Power (Faul et al., 2009) revealed that, given an alpha of .05 and conventionally assumed power of .80, a sample size of 159 participants would be sufficient to observe a medium effect size ($f = .25$). We decided to increase the sample size to 180 to compensate for possible withdrawals. The participants (who were from the United States) were recruited via the Internet platform Prolific (136 females, 44 males, $M_{age} = 32.79$, $SD =$

12.23). In line with the preregistration, responses from 12 participants who completed the research using mobile phones were excluded from the analysis. Participants were remunerated ($1.65), and they took part voluntarily. Before beginning the experiment, participants gave their informed consent and were informed about the general purpose of the study. They were also told that they could withdraw at any point. The study was designed and conducted in accordance with guidelines enforced by the University Ethical Committee and preregistered at the Open Science Framework (https://doi.org/10.17605/OSF.IO/3E65N).

Procedure

The study was conducted online. Participants were randomly assigned to one of three experimental conditions: (a) negative mental imagery (n = 63); (b) positive mental imagery (n = 54); and (c) the control condition (n = 49). For the positive and negative mental imagery conditions, participants were asked to imagine, respectively, either positive or negative circumstances of engaging in a risky action presented to them. In the control condition, the participants' task was to freely imagine circumstances related to risk taking without specifying the valence of mental images. The control condition in Study 2, therefore, mirrored that of Study 1.

As with Study 1, Study 2 comprised five phases. Participants read randomly displayed descriptions of different real-life actions involving risk (Blais & Weber, 2006) (e.g., "Not returning a wallet you found that contains $200" for the ethical activity; Table 2 shows each of the risky activities included in the study). The participants' task was to imagine themselves being actively involved in the risky activity in their mind's eye and provide up to three descriptions of their mental images. Next, they were asked to rate the valence of each mental image they reported ("What was your image like?") on a scale ranging from 0 – *very negative* to 100 – *very positive*. Finally, participants answered questions about their willingness to take risky action ("Would you be willing to take action in this situation?") on a scale ranging from

1 – *definitely no* to 7 – *definitely yes* (with higher numbers indicating preparedness to take more risks). Before the start of each phase, participants were reminded of the type of mental imagery they should generate and the need for it to be dependent on the condition to which they were assigned.

Results

Qualitative Analysis of the Content of Entries Provided by Participants

As with Study 1, before testing the preregistered hypothesis, we invited three independent judges to analyze the content of descriptions provided by participants. The purpose and procedure of this qualitative analysis followed the same assumptions and methods as those used in Study 1.

We collected a total number of 2663 participants' entries relating to risky actions. The independent judges agreed that in 2177 cases (82% of all entries), participants provided descriptions showing that they had generated visual mental images of risky situations presented to them. The descriptions evaluated as "visual imagery" included statements such as: "A nice forest scene, an assortment of tents, and a campfire. Maybe near a lake. I'm there with friends, and this is something we all decided to do together"; "Seeing myself and this person enjoying dinner and a movie"; and "Images of men screaming on the trading floor and me standing there wondering what the hell everyone is screaming for." (see Supplementary Materials for additional examples of entries provided by participants in Study 2).

Manipulation Check

As in Study 1, participants' responses in each group were averaged over all activities to create general indices of the mean valence of mental imagery (Cronbach's α = .859) and the mean willingness to take risky actions (Cronbach's α = .449) for each participant separately (see detailed descriptive statistics for each activity and condition in Table 2). In line with our preregistration, we excluded 19 entries where participants did not provide any descriptions of

mental images or provided descriptions entirely unrelated to the task (as indicated in the response box). Again, in the first step of the analysis, we used responses provided by all participants without excluding entries evaluated by independent judges as "non-imagery."

First, we tested the effectiveness of our manipulation by comparing mean participants' evaluations of the valence of mental images they generated across the three conditions. The results of a one-way analysis of variance (ANOVA) indicated that the valence of mental imagery reported by participants differed significantly across conditions: $F(2, 163) = 114.237$, $p < .001$, $\eta^2 = .584$. The highest ratings were found in the positive mental imagery condition ($M = 67.787$, $SD = 16.498$), the medium ratings in the control condition ($M = 49.436$, $SD = 12.771$), and the lowest ratings in the negative mental imagery condition ($M = 23.278$, $SD = 17.788$). Further post hoc comparisons using Tukey's test showed significant differences in the valence ratings between all pairs of conditions: positive vs. negative mental imagery: $t(115) = -14.974$, $p < .001$, Cohen's $d = -2.587$; negative mental imagery vs. controls: $t(110) = 8.568$; $p < .001$, Cohen's $d = 1.656$; and positive mental imagery vs. controls: $t(101) = -5.803$, $p < .001$, Cohen's $d = -1.236$). In both experimental conditions, the valence ratings deviated significantly from 50 (the middle of the scale) and in the direction presumed by the manipulation: positive mental images: $t(53) = 7.923$, $p < .001$, Cohen's $d = 1.078$, 95% CI [0.739, 1.411]; and negative mental images: $t(62) = -11.923$, $p < .001$, Cohen's $d = -1.502$, 95% CI [-1.861, -1.138]). The valence of mental imagery in the control condition did not deviate significantly from 50: $t(48) = -.309$, $p = .758$, Cohen's $d = -0.044$, 95% CI [-0.324, 0.236]. The analysis therefore confirmed that the manipulation of the valence of mental imagery used in Study 2 was successful.

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Table 2

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Hypothesis Testing

To test the main hypothesis, we conducted two preregistered analyses. First, we compared participants' willingness to engage in risky actions between the three conditions. As expected, the results of a one-way ANOVA showed a main effect of our manipulation (F[2, 163] = 7.979, $p < .001$, $\eta^2 = .089$), with the greatest willingness to take risky actions in the positive mental imagery condition (M = 4.378; SD = 0.954); the lowest willingness to take risky actions in the negative mental imagery condition (M = 3.55; SD = 1.313); and a medium willingness to take risky actions in the control condition (M = 3.927; SD = 1.001). A post hoc Tukey test indicated that the difference in the willingness to take risks between positive and negative imagery conditions was significant: $t(115) = -3.995$, $p < .001$, Cohen's $d = -0.712$. The comparisons between the experimental and control conditions were not significant. Negative mental imagery vs. controls: $t(110) = 1.768$, $p = .184$, Cohen's $d = 0.317$; and positive mental imagery vs. controls: $t(101) = -2.047$, $p = .104$, Cohen's $d = -0.462$.

We then examined correlations between the valence of mental imagery and the willingness to take risks. We observed significant and positive correlations between the valence of mental imagery and the willingness to engage in risky actions in each condition but with varying strengths. In the conditions where participants generated either positive or negative mental images, the relationships were moderate, respectively: $r(52) = .515$, $p < .001$, 95% CI [.327, 1.000] and $r(61) = .299$, $p = .009$, 95% CI [.096, 1.000]. The correlation in the control condition was also significant: $r(47) = .704$, $p < .001$, 95% CI [.560, 1.000]. We compared all possible pairs of correlation coefficients and found that the comparison between the negative imagery condition and the control condition was the only significant one (Fisher's $z = -2.9021$, $p = .001$). The significance of the remaining correlations was beyond the commonly assumed threshold (positive imagery vs. control condition: $p = 0.06$, positive imagery vs. negative imagery: $p = 0.08$).

As was the case in Study 1, we conducted an additional, non-preregistered analysis of the valence of mental images and the willingness to take action in risk using only those entries that were coded as "visual imagery" in the qualitative analysis. After excluding 486 separate entries and data units related to them (i.e., ratings of the valence of mental images and the willingness to take risky actions) rated by independent judges as "non-images," we carried out a quantitative analysis using only the data assigned by the judges to the category of "visual mental images." First, we repeated the manipulation check by comparing the mean valence of mental images reported by participants in each condition. The ANOVA results revealed a significant difference in the mean valence of mental images: $F(2, 163) = 98.339$, $p < .001$, $\eta^2 = .547$. The highest ratings were in the positive mental imagery condition ($M = 67.696$, $SD = 17.884$); the medium ratings in the control condition ($M = 50.248$, $SD = 12.875$); and the lowest ratings in the negative mental imagery condition ($M = 24.036$, $SD = 18.871$). This provided further evidence that the manipulation was successful. Post hoc Tukey test comparisons between all pairs showed significant differences in the mean valence of mental images across all conditions (Table 3). Next, we compared the declared mean willingness to take risks across the three conditions. The ANOVA results again showed a significant effect of our manipulation ($F[2, 163] = 6.773$, $p < .001$, $\eta^2 = .077$), with the greatest willingness to take risky actions in the positive mental imagery condition ($M = 4.313$, $SD = 1.016$); a medium willingness to take risks in the control condition ($M = 3.979$, $SD = 0.974$); and the lowest willingness to take risks in the negative mental imagery condition ($M = 3.557$, $SD = 1.282$). Post hoc Tukey test comparisons between all pairs of conditions replicated the results from the quantitative analysis, revealing a significant difference in the mean willingness to take risky actions between positive and negative mental imagery conditions (Table 3). Moreover, the mean valence of mental images in each experimental condition was significantly different from the middle of the scale (50), in the direction assumed by the

manipulation. Positive mental images: $t(53) = 7.271$, $p < .001$, Cohen's $d = 0.989$, 95% CI [0.660, 1.313]; negative mental images: $t(62) = -10.921$, $p < .001$, Cohen's $d = -1.376$, 95% CI [-1.718, -1.027]. In the control condition, the mean valence did not differ from the middle of the scale: $t(48) = -0.135$, $p = .893$, Cohen's $d = -0.019$, 95% CI [-0.257, 0.294]. These results supported the hypothesis that there would be a causal link between the valence of mental images and the propensity to take risky actions.

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Table 3

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We calculated correlation coefficients for the relationship between the valence of mental images and declared readiness to take risky actions. In the conditions with positive and negative images, the correlation was moderate and significant, respectively: $r(52) = .579$, $p < .001$, 95% CI [.406, 1.000] and $r(61) = .353$, $p = .002$, 95% CI [.155, 1.000]. The correlation observed in the control condition was also significant: $r(47) = .613$, $p < .001$, 95% CI [.439, 1.000]. Finally, we compared all possible pairs of correlation coefficients; only the difference between the negative imagery condition and the control condition was significant (Fisher's $z = -1.759$, $p = .03$). In the remaining comparisons, the significance level did not reach the commonly assumed threshold: positive imagery vs. control condition: $p = .30$, positive imagery vs. negative imagery: $p = .06$. The analysis, therefore, mirrored the results from the entire data set.

Similarly to Study 1, to address potential variability between risky activities, we fitted a multilevel linear regression model predicting participants' willingness to engage in risky activities with the condition, ratings of valence (mean-centered in each condition), and their interaction as fixed effects, and participants and risky activities as random intercept effects. The results showed a main effect of our manipulation $R^2 = .479$, with the greatest willingness

to take risky actions in the positive mental imagery condition (M = 4.371; 95% CI [3.859, 4.882]); the lowest willingness to take risky actions in the negative mental imagery condition (M = 3.519; 95% CI [3.017, 4.021]); and a medium willingness to take risky actions in the control condition (M = 3.928; 95% CI [3.410, 4.447]). A post hoc Tukey test indicated that the difference in the willingness to take risks between positive and negative imagery conditions was significant: $t(167.56) = 4.636$, $p < .001$. The comparisons between the experimental and control conditions were not significant: negative mental imagery vs. control condition, $t(167.56) = 2.168$, $p = .079$, and positive mental imagery vs. control condition, $t(167.56) = -2.265$, $p = .064$ (Figure 2).

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Figure 2

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Next, we examined relationships between the valence of mental imagery and the willingness to take risks. We observed significant and positive associations between these variables in each condition: positive imagery condition, $b = 0.050$, $p < .001$, 95% CI [0.042, 0.058], negative imagery condition, $b = 0.036$, $p < .001$, 95% CI [0.027, 0.044], and control condition, $b = 0.054$, $p < .001$, 95% CI [0.046, 0.062]. The slopes did not differ significantly between conditions. The pattern of findings based on multilevel modeling remained the same when entries rated by independent judges as “non-images” were excluded.

Summary

The results from Study 2 supported our prediction that the valence of mental imagery would be positively related to the willingness to take risks. Moreover, the relationship between the positivity or negativity of mental images and the proneness to engage in risky actions was causal. Preparedness to take risks was greater in the positive imagery group than in the negative imagery group. Nevertheless, contrary to the results of Study 1, we found that

in the control condition, the mean valence of mental images did not differ significantly from 50. We assume that this discrepancy in results could be driven by the characteristics of risky activities (see Tables 1 & 2) or samples. In particular, in Study 2, most participants were women. It means that the effect we found here should be replicated in a sample with an even gender balance and on a larger set of risky activities.

Study 3

The results of Studies 1 and 2 supported our expectation that visual mental imagery depicting circumstances of risky actions would be associated with participants' declared willingness to take risks. The potential limitation of these two studies was that participants were explicitly instructed to generate mental images when faced with risky situations. The aim of Study 3 was to investigate whether decision makers, when faced with a dilemma of whether to take a risky action or not, would spontaneously produce mental images without being directly asked to do so. This would indicate that people engage in a spontaneous generation of mental images, thus supporting our general thesis that visual mental imagery is used as a guide for making decisions that entail risk.

For Study 3, we assumed that participants might access different types of information arising from different sources and originating in diverse psychological processes, such as logical arguments concerning threats and benefits, feelings and sensations, memories, and mental images (Weber et al., 1995; Weber & Lindemann, 2007, Zaleskiewicz et al., 2023). We aimed to investigate whether participants would state that they based their decisions on mental images besides other types of information. To simplify the presentation of our methods and results, we hereafter use the term *decision inputs* when referring to said information.

Study 3 was exploratory, that is, we did not test a specific hypothesis. We did, however, predict that participants would use mental images in addressing risky dilemmas and that there

would be a positive correlation between the valence of decision inputs (including mental images) and a declared willingness to take risks.

Method

Participants

One-hundred-and-fifty participants from the United States were recruited via the Internet platform Prolific (75 females, 75 males, M_{age} = 37.70, *SD* = 13.52). Since this was an exploratory study, we determined the sample size using a heuristic sample selection, considering our budgetary restrictions (Lakens, 2022). In line with the preregistration, data from six participants who used mobile phones were excluded from the analysis. Participants, all of whom were volunteers, were remunerated ($0.82). Before the study began, participants signed an informed consent form that briefly described the general purpose of the study; they were also told that they could withdraw at any point. The study was designed and conducted following ethical guidelines enforced by the University Ethical Committee and preregistered at the Open Science Framework (https://doi.org/10.17605/OSF.IO/KWCBJ).

Procedure

The study was conducted online. As with Studies 1 and 2, Study 3 comprised five steps. In each one, participants read descriptions of risky activities (presented in random order) taken from Study 2. Participants' task was to assume that they were engaged in risky activities presented to them and write down the first thing that crossed their minds. They then rated the valence of their descriptions ("What was the thing you just described like?") on a scale ranging from 0 (*very negative*) to 100 (*very positive*) and declared their willingness to take action (e.g., "Would you be willing to take this action [i.e., not returning a wallet][2]?") on a 7-

[2] For each situation, the question measuring the dependent variable included the description of the specific risky situation to remind participants.

point scale with responses ranging from 1 (*definitely no*) to 7 (*definitely yes*). In the final phase, participants were presented with each of the descriptions they provided in the previous phases of the research. Their task was to classify each description into only one of the following categories: “Logical argument,” “Feeling and sensation,” “Memory,” “Visual imagery,” and “Other.” The categories were defined for the participants and selected based on studies of the different types of decision strategies people adopt when dealing with real-life dilemmas (Weber et al., 1995; Weber & Lindemann, 2007; Zaleskiewicz et al., 2023).

Results

First, we calculated the overall frequency (i.e., aggregated over all risky activities) with which participants used different decision inputs. The multinomial test revealed significant differences between the frequency of reported decision inputs, $\chi^2(4) = 355.653$, $p < .001$. Taken together, participants’ choices were most often guided by logical arguments, followed by feelings and sensations, visual images, and memories, in descending order (Table 4). Overall, mental images were indicated as decision inputs in 16% of cases. Post-hoc pairwise comparisons with Bonferroni correction showed that there were no statistical differences between the frequency of reporting logical arguments and feelings and sensations ($p = .479$) as well as visual images and memory ($p = .146$). All other comparisons were statistically significant ($p < .001$).

Next, we analyzed the relative frequency of decision inputs across five risky activities (financial, ethical, social, health, and recreational), as shown in Figure 3.

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Table 4

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Figure 3

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As can be seen in Figure 3, the frequency of using decision inputs differed among risky activities, with logical arguments predominant in the financial and health activity, feelings and sensations in the ethical activity, and visual images in the recreational activity. In the social activity, participants declared using logical arguments and feelings or sensations with almost equal relative frequency.

Since we intended to focus on the declared frequency of using mental images, we compared responses concerning this specific decision input among different risky activities. In other words, we made comparisons among activities of how frequently mental images were used as decision inputs within these activities. Table 5 shows that mental images occurred most frequently in recreational activity—56% of all mental images reported as a decision input were generated in response to recreational action. In other activities, mental images were used relatively less frequently. The multinomial test indicated significant differences between activities, $\chi^2(4) = 97.130$, $p < .001$. Post-hoc pairwise comparisons with Bonferroni correction clearly showed that visual images were reported more frequently in recreational activity compared to activities in other domains (all *p*s < .001). We found no statistically significant differences in using visual mental images as decision inputs between financial, ethical, social, and health activities.

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Table 5

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To test our hypothesis that there would be a positive relationship between the valence of decision inputs and a declared willingness to take risky actions, we conducted a series of

correlation analyses. We found strong and positive correlations between the valence of all decision inputs and the declared willingness to take risks (Table 6; see also Table S1 in Supplementary Materials for correlations within five separate risky activities). No significant differences were found in the strength of compared pairs of correlation coefficients.

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Table 6

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To account for variability between risky activities (Cronbach's α for the ratings of valence and willingness to take risk were 0.211 and 0.411, respectively), we ran a multilevel regression model with participants random intercept effect and the valence of mental images as random slope varying across risky activities. In this model, we introduced the interaction between the valence of mental images and reported decision input (i.e., visual images vs. other inputs) as fixed factors. The relationship between the valence of mental images and willingness to take risks was significant, $b = 0.047$, $t(5.713) = 10.177$, $p < .001$, 95% CI [0.036, 0.058], $\delta t = 0.037$ (Fig. 4). The effects of decision input and the interaction of this factor with the valence of mental imagery were not significant ($ps > .100$).

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Figure 4

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Summary

In conclusion, we observed that participants used a range of decision inputs, the perceived valence of which was significantly associated with a declared willingness to take risky actions. From our theoretical perspective, mental images served as a spontaneous decision input in 16% of all cases, with the highest prevalence in recreational activity (56%)

compared to other activities. This suggests that envisioning the future using visual imagery may be considered a “mental tool” in the decision-making process in real-life situations.

To investigate whether the effects found in Study 3 and in two previous studies could have been driven by some basic characteristics of the activities that were presented to participants, we conducted an auxiliary study with an independent sample (the detailed results of the auxiliary study are presented in Supplementary Materials). We asked participants to judge the five risky activities on two scales measuring imaginability and concreteness because these two characteristics can impact the easiness with which people generate mental imagery (Paivio, 1969). Suppose the activities we used in this project differed remarkably in how imaginable and concrete they were in the eyes of the participants. In that case, such differences might explain why mental imagery was more or less frequently indicated as a decision input in Study 3. The results of the auxiliary study showed that (1) the mean ratings of the risky activities on both scales were above the midpoint of the scales (i.e., 4) and (2) the activity in the recreational domain, for which mental imagery was most often indicated as a decision input in Study 3, had the highest rating in imaginability and relatively high rating in concreteness. The former effect suggests that all activities had a fair potential to be visualized. The latter effect indicates that risky activities in some domains (e.g., recreational domain) may have more significant imaginative potential than activities in other domains (e.g., financial domain). Undoubtedly, this result needs further examination.

General Discussion

The present research provided initial evidence for the role of visual mental imagery in people’s willingness to engage in risk-taking. First, we showed that when instructed, participants were able to produce mental images of risky actions. The mean valence of mental images generated by participants in Study 1 was negative rather than positive, but this effect was not replicated in Study 2, which suggests that it requires further investigation. Second, we

found that the valence of mental images was positively correlated (r = .29 to r = .70) with the willingness to engage in risky activities. It means that generating more positive mental images was associated with people's higher propensity to take risks. Third, in Study 2, we provided evidence that the valence of visual mental images could be successfully manipulated, and our instructions affected not only the valence of mental images but also the willingness to engage in risky actions. Finally, we found that participants declared using mental imagery as a decision input (i.e., a source of information that helped them make choices) even when they were not instructed to do so. This suggests that participants were able to use mental imagery spontaneously in their choices.

The results from all three studies show that episodic simulations of the future based on activating mental imagery were involved in participants' simulations of their potential engagement in risky activities they were presented with. Previous studies have suggested that mental images are associated with risk perception (Leiserowitz, 2006; Sinclair et al., 2021; Smith & Leiserowitz, 2012; Sobkow et al., 2016; Zaleskiewicz, Traczyk et al., 2023). The outcomes of our current research indicate that imaginative visualizations can also correlate with or causally influence the willingness to engage in risky activities. This suggests that mental imagery that is involved in the decision-making process not only regulates the way people evaluate threats but also may be linked to people's motivation to either approach risk or avoid it. Indeed, as previous studies have documented, mental imagery can operate as a "motivational amplifier" (Renner et al., 2019), that is, when people generate positive images of reward activities, their motivation to engage in such activities increases (Ji et al., 2021). Our results support such a conclusion and additionally indicate that when imagery-based episodic simulations are negative in valence, they encourage restraint. In the present instance, mental imagery both amplified and inhibited certain behaviors. Future research might investigate whether the primary role of mental imagery generated in decision making is to

strengthen the motivation to engage in risky activities to achieve some benefits or the motivation to restrain from risky actions to avoid potential losses.

One issue that might be discussed further is whether the link between the valence of mental imagery and a propensity for risk taking might be mediated by other factors. Given that mental imagery is closely related to affect (Blackwell, 2020; Holmes & Mathews, 2005; Ji et al., 2016; Nanay, 2023), we would suggest that emotions could be considered as a candidate, particularly because feelings play an essential role in both risk perception and risk taking (Damasio, 2005; Loewenstein et al., 2001; Slovic et al., 2004, 2007). However, bearing in mind that affect might be automatically evoked when people are faced with risk (Damasio, 2005; Peters & Slovic, 2007), it is also possible that emotions direct the search for imagery in memory (see also Blackwell, 2019; Holmes et al., 2016; Nanay, 2023). In such a case, mental images could either strengthen or disrupt the impact of affect on risk perception or risk-taking propensity. Undoubtedly, future research on imagery-based decision making should carefully investigate the nature of reciprocal relationships between mental imagery, emotions, and decision making.

Another finding in Study 1 was that mental images evoked in response to risky situations were largely negative. This is perhaps not surprising because risky situations, by definition, can have negative repercussions, and negative outcomes are weighted more heavily than positive outcomes (Baumeister et al., 2001; Kahneman & Tversky, 1979). Being alert to potentially harmful events is an evolutionary adaptation. Moreover, previous studies have found a negative valence of mental images spontaneously generated by participants in response to risky situations to be prevalent (e.g., Sobkow et al., 2016; Traczyk et al., 2015). It is possible that one highly adaptive function of decision-related episodic simulations may be designed to protect the decision maker against exceeding the limits of acceptable risk (just as

somatic markers do; Damasio, 2005). However, this is a speculative idea based only on the results of our Study 1. Therefore, it requires further investigation.

The results of the present research (Study 2) also point to a possible causal relationship between the valence of mental images and the tendency to engage in risky actions. In particular, participants who were instructed to imagine positive situations in which they engaged in risky behaviors not only declared more positive mental images (which indicated that our manipulation was successful) but were also more willing to accept risk. We would argue that the possibility of inducing the specific valence of mental images that people process further in their decision making might have practical implications. For example, instructing people to imagine the positive consequences of an extremely stressful and uncertain event (e.g., the COVID-19 pandemic) may increase positive feelings (Petrova et al., 2022; Sobkow et al., 2020) and well-being. Relatedly, encouraging people to imagine the negative aspects of a risky action may encourage them to be more risk averse and more prepared to take protective measures (Zaleskiewicz et al., 2020). Rakow et al. (2015) stressed the proper design of emotional information as one of the main challenges in risk communication. Bearing in mind that mental imagery has strong connections with emotions (Blackwell, 2020; Holmes & Mathews, 2005; Ji et al., 2016), it may be expected that mentally vivid and affect-laden risk communication would be particularly important in the context of threats that are underestimated and difficult to imagine spontaneously (e.g., those relating to climate change; Sundblad et al., 2007).

Studies have shown that future cognitions can appear involuntarily without external cues (Berntsen, 2019) and that people can use mental images spontaneously in planning and decision making (Barsics et al., 2016; D'Argembeau et al., 2011). We would suggest that mental imagery might be driven not only by various external factors (e.g., instructions) but might also emerge spontaneously in the decision-making process. As was noted above, we

tested this hypothesis in Study 3 and found that in approximately 16% of risky situations, participants declared that visual images appeared spontaneously in their minds and that they used them as decision inputs. Moreover, the valence of these visual mental images was highly correlated ($r > .80$) with the willingness to take risky actions. It should be noted, however, that, in this study, participants could classify their descriptions (decision inputs) into only one category ("Logical argument," "Feeling and sensation," "Memory," "Visual imagery," and "Other"). It is, therefore, possible that some participants who chose one of the non-imagery categories had also generated a visual mental image that evoked a specific emotion or was linked to memory. If feeling or memory was more salient, it was reported instead of a mental image. It is also plausible that some experiences in the form of mental images were rapidly recoded in another format (e.g., a logical argument) that was easier to report. Taken together, there are reasons to presume that the actual frequency of mental images used by participants in the form of decision inputs was higher than 16%. Moreover, as already mentioned earlier in this paper, even when people are prompted to use verbal processing, visual mental imagery is also spontaneously activated (Amit et al., 2017).

We also found substantial differences in how often participants declared that mental images served as an input in their decision making across the risky activities. For example, in recreational activity, the category of mental images was most often referred to as a source of information. By contrast, mental imagery was trumped by other decision inputs in the financial activity (with logical arguments being the most prominent). It should be noted, however, that investigating differences across domains was not the aim of the present study, so we used only one activity per risk domain. Nevertheless, the results were sufficiently interesting to merit further examination. Previous studies have demonstrated substantial differences among risk domains in the degree to which diverse psychological processes are involved in risk perception and risk taking (Hanoch et al., 2006; Weber et al., 2002), and the

results from Study 3 bore this out. One of the sources of such heterogeneity may be the use of different decision inputs (e.g., a stronger or weaker tendency to use mental images).

Limitations and Future Research Directions

The present study has several limitations. One major methodological challenge faced by researchers studying mental imagery is the accurate identification and reliable measurement of the content processed by participants. In other words, even if participants are instructed to generate mental images, researchers should find ways to document that what is generated can be reliably identified as visualization. In the present study, we asked participants to report their mental images, and independent judges subsequently coded them. We found that the vast majority of participants' entries were coded as visual imagery, suggesting that (even though the experiments were conducted online), participants followed our instructions and did generate mental images. Nevertheless, future studies should employ indirect measures of assessing mental imagery (e.g., neuroimaging techniques) to provide stronger support for the conclusion that mental images of risk appear spontaneously in people's minds.

Another limitation that may be a promising line of future research concerns the fine-grained recording of affective/emotional responses to risk evoked by mental imagery. In the present study, the valence of mental images was rated sequentially for each scenario. Even though such an approach is straightforward (and has been employed successfully in studies on affect and decision making; Asutay et al., 2019; Asutay & Västfjäll, 2022), it misses essential aspects regarding the temporal dynamics of mental imagery. For instance, it is unclear whether participants based their risk preference on a global evaluation of the mental image valence, used the emotional peak of the most vivid mental image, or used the valence of the last image. This problem might be tackled by adopting classic methods of studying subjective emotional experience (Fredrickson & Kahneman, 1993), which would allow for a more

detailed investigation of the dynamics of mental image generation and possibly reveal a link between subjectively experienced emotions elicited by mental imagery and risk-taking.

The third limitation refers to the design of our Studies 1 and 2. In both studies, we explicitly asked participants to generate mental imagery. In other words, we did not have a non-imagery condition in which some participants would be requested to think about a decision situation differently (e.g., to consider analytical arguments). Alternatively, in a non-imagery condition, participants could be engaged in a mental activity unrelated to the decision-making process, which could block or disrupt visual mental imagery before making a decision. Future studies should use the non-imagery condition to more accurately detect those effects that are specific to visual mental imagery.

The fourth limitation refers to the fact that, in our present research, we asked participants to visualize their engagement in different risky actions. Our prediction was that the valence of such visualizations could be helpful in deciding whether to take a risky action or not. However, in many real-life situations, people are faced with a choice between two distinct alternatives, of which one is safe and another risky. In such cases, decision makers can generate independent mental images illustrating both alternatives and, afterward, make a choice based on comparing the valence and vividness of both images. Future research might use a design in which participants produce not only images of a risky action but images of all alternatives at hand.

Finally, our research has shown that mental imagery plays an important causal role in risky decision making. However, besides the abovementioned limitations and possible heterogeneity among risky activities, we argue that these effects may be moderated by some dispositional characteristics. In particular, people may differ in how often they spontaneously use mental imagery in real life (e.g., Zaleskiewicz et al., 2023) and how vivid their mental images are (Campos & Perez-Fabello, 2009). For example, in the extreme case of people with

aphantasia (Keogh & Pearson, 2018) who do not generate visual mental images, it could be hypothesized that they base their risky decisions on logical inputs and calculations more often than people from the general population. Future research should investigate this issue and include individual differences measures as potential moderators.

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Table 1

Descriptive statistics for risky activities used in Study 1

Risky situations	Valence of mental imagery		Willingness to take risks	
	M	*SD*	*M*	*SD*
Taking strong medications without consulting your doctor	25.294	20.492	2.462	1.790
Forging your partner's signature on a legal document	28.761	20.770	2.624	1.977
Betting a year's salary on sports	23.659	20.016	1.471	1.267
Having casual unprotected sex	41.743	27.872	3.362	2.255
Driving fast on a winding road in the mountains	41.280	29.187	3.308	2.287
Average ratings for all situations	32.147	23.667	2.645	1.915

Note: For Valence of mental imagery: min = 0; max = 100. For Willingness to take risk: min = 1; max = 7.

Table 2

Descriptive statistics for risky activities used in Study 2

Risky situations	Positive mental imagery condition				Negative mental imagery condition				Control condition			
	Valence of mental imagery		Willingness to take risk		Valence of mental imagery		Willingness to take risk		Valence of mental imagery		Willingness to take risks	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Not returning a wallet you found that contains $200	59.45	23.336	2.83	2.045	21.194	21.317	2.794	2.025	36.520	26.495	2.653	1.809
Investing 10% of your annual income in a moderately growth-diversified fund	73.731	16.94	5.815	1.199	31.232	25.851	4.387	1.841	58.501	19.080	4.592	1.743
Engaging in unprotected sex	59.09	27.678	3.722	2.302	15.804	18.618	3.048	2.113	40.122	27.744	3.408	2.432
Camping in the wilderness	79.445	21.508	5.481	1.634	23.982	28.177	4.000	2.195	64.025	24.779	5.245	1.809
Dating someone you’re working with	66.625	28.261	3.981	2.159	24.293	19.060	3.444	1.990	48.011	23.535	3.735	2.243
Average ratings over all situations	67.787	16.498	4.378	0.955	23.278	17.788	3.550	1.313	49.436	12.771	3.927	0.955

Note: For Valence of mental imagery: min = 0; max = 100. For Willingness to take risks: min = 1; max = 7

Table 3

Detailed results for post hoc comparisons between pairs of conditions after qualitative analysis

Conditions compared	*t*	*p*	Cohen's *d*
	Valence of mental images		
Positive imagery vs. negative imagery	*t*(115) = -13.866	< .001	- 2.370
Positive imagery vs. control condition	*t*(101) = -5.208	< .001	-1.111
Negative imagery vs. control condition	*t*(110) = 8.105	< .001	1.586
	Declared willingness to take risks		
Positive imagery vs. negative imagery	*t*(115) = -3.660	< .001	-0.648
Positive imagery vs. control condition	*t*(101) = -1.521	.284	-0.335
Negative imagery vs. control condition	*t*(110) = 1.988	.118	0.364

Table 4

Overall frequencies of decision inputs declared in Study 3

Decision input	Frequency of occurrence	Percentage of occurrence	95% CI	
			LL	UL
Logical arguments	282	39.2%	35.6%	42.8%
Feelings and sensations	236	32.8%	29.4%	36.3%
Visual imagery	115	16.0%	13.4%	18.9%
Memories	80	11.1%	8.9%	13.6%
Other	7	1.0%	0.4%	2.0%

Note. Confidence intervals are based on independent binomial distributions.

Table 5

Percentage of reported mental images as decision inputs across risk domains in Study 3

Decision domain	Frequency of occurrence	Percentage of mental images	95% CI	
			LL	UL
Ethical	14	12.2%	6.8%	19.6%
Financial	8	7.0%	3.1%	13.2%
Health	13	11.3%	6.2%	18.6%
Social	15	13.0%	7.5%	20.6%
Recreational	65	56.5%	47.0%	65.7%

Note. Confidence intervals are based on independent binomial distributions.

Table 6

Pearson's correlations between the participants' evaluations of the valence of decision inputs and their willingness to take risky activities

Decision strategy	Person's *r*	*p* value	95% CI
Logical arguments	$r(280) = .774$	< .001	[.723, .816]
Feelings and sensations	$r(234) = .824$	< .001	[.780, .860]
Visual images	$r(113) = .801$	< .001	[.726, .858]
Memories	$r(78) = .818$	< .001	[.731, .879]
Other	$r(5) = .386$	.450	[-.620, .912]

Figure 1

The predicted valence of mental images (panel A), willingness to take risk (panel B), and the relationship between these variables (panel C) across five risky activities (treated in the model as a random intercept effect). Solid and dotted lines represent predictions based on all data and content categorized by independent judges as visual images, respectively. Error bars represent 95% confidence interval

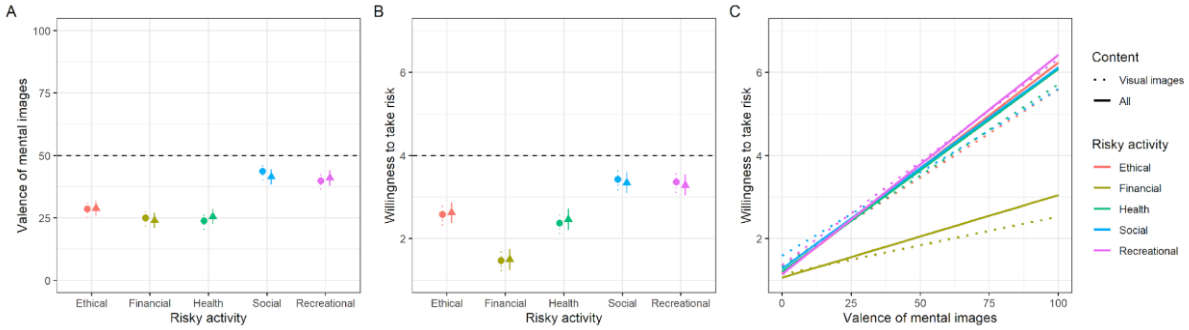


Figure 2

The predicted valence of mental images (panel A), willingness to take risk (panel B), and the relationship between these variables (panel C) across three experimental conditions. Participants and risky activities were treated in the model as random intercept effects. Solid and dotted lines represent predictions based on all data and content categorized by independent judges as visual images, respectively. Error bars represent 95% confidence interval

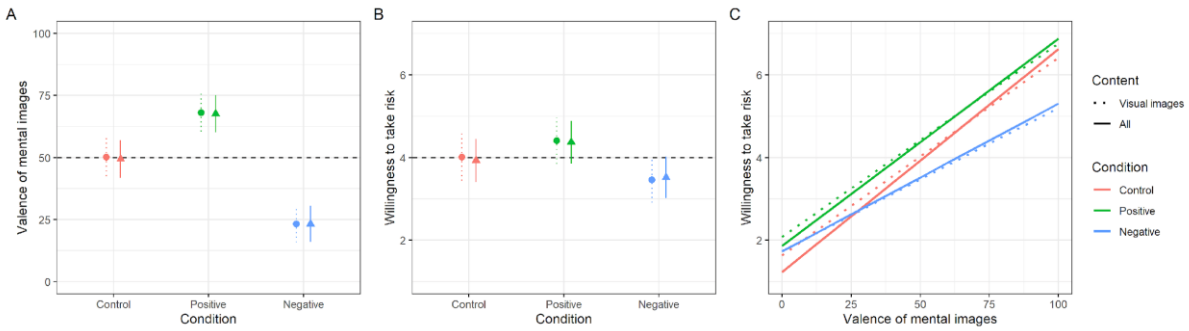


Figure 3

Proportions of five categories of decision inputs among risky activities in Study 3. Error bars represent 95% confidence interval

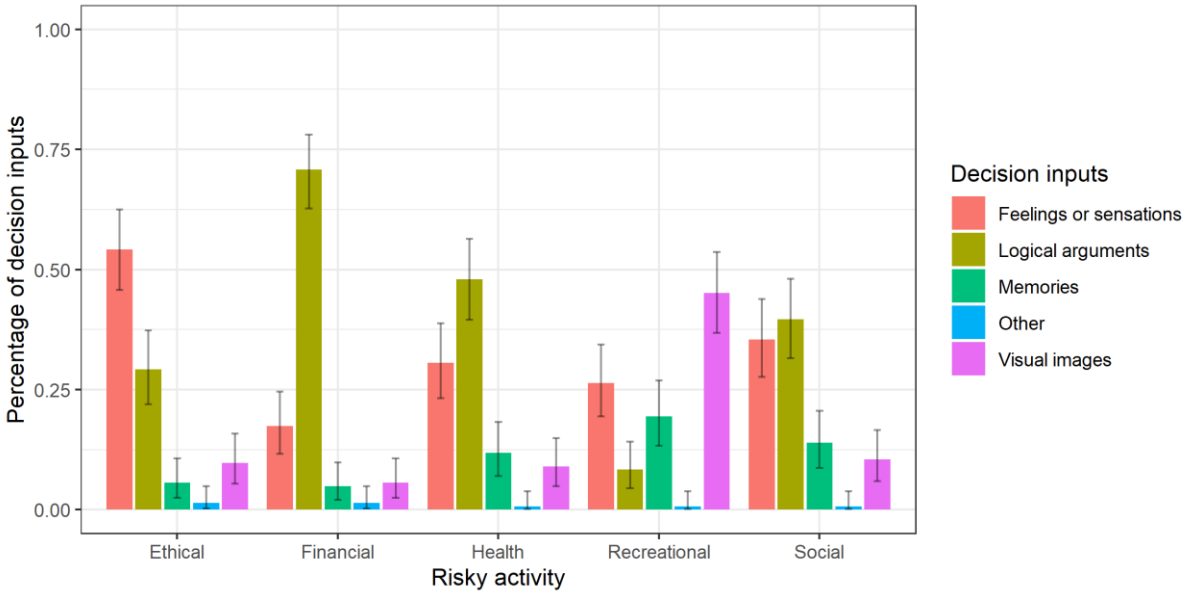


Figure 4

Predicted effect of the valence of mental images on willingness to take risk as a function of decision input: visual images (dotted lines) vs. other decision inputs (solid lines) across five risky activities.

